

Exercise Set 10.1 (Page 463)

Evaluate (Problems 1 – 19):

$$1. \int_0^1 \int_1^2 dx \, dy$$

$$\begin{aligned} \text{Sol. } \int_0^1 \int_1^2 dx \, dy &= \int_0^1 [x]_1^2 dy = \int_0^1 [2 - 1] dy \\ &= \int_0^1 dy = [y]_0^1 = 1 - 0 = 1. \end{aligned}$$

$$2. \int_1^2 \int_0^3 (x + y) dx \, dy$$

$$\begin{aligned} \text{Sol. } \int_1^2 \int_0^3 (x + y) dx \, dy &= \int_1^2 \left(\left[\frac{x^2}{2} \right]_0^3 + [xy]_0^3 \right) dy \\ &= \int_1^2 \left[\left(\frac{9}{2} - \frac{0}{2} \right) + (3y - 0y) \right] dy \\ &= \int_1^2 \left[\frac{9}{2} + 3y \right] dy = \frac{9}{2} [y]_1^2 + 3 \left[\frac{y^2}{2} \right]_1^2 \\ &= \frac{9}{2} (2 - 1) + \frac{3}{2} (4 - 1) = 9. \end{aligned}$$

$$\int_2^4 \int_1^2 (x^2 + y^2) dy \, dx$$

$$\begin{aligned}
 \text{Sol. } \int_2^4 \int_1^2 (x^2 + y^2) dy dx &= \int_2^4 \left[x^2 y + \frac{y^3}{3} \right]_1^2 dx \\
 &= \left[\int_2^4 x^2 (2-1) + \frac{1}{3}(8-1) \right] dx \\
 &= \int_2^4 \left[x^2 + \frac{7}{3} \right] dx = \left[\frac{x^3}{3} + \frac{7}{3}x \right]_2^4 \\
 &= \frac{1}{3}(64-8) + \frac{7}{3}(4-2) = \frac{70}{3}.
 \end{aligned}$$

$$4. \int_0^1 \int_{x^2}^x xy^2 dy dx$$

$$\begin{aligned}
 \text{Sol. } \int_0^1 \int_{x^2}^x xy^2 dy dx &= \int_0^1 \left(x \left[\frac{y^3}{3} \right]_{x^2}^x \right) dx = \int_0^1 \left[x \left(\frac{x^3}{3} - \frac{x^6}{3} \right) \right] dx \\
 &= \int_0^1 \left(\frac{x^4}{3} - \frac{x^7}{3} \right) dx = \left[\frac{x^5}{15} - \frac{x^8}{24} \right]_0^1 \\
 &= \left(\frac{1}{15} - \frac{0}{15} \right) - \left(\frac{1}{24} - \frac{0}{24} \right) = \frac{1}{40}.
 \end{aligned}$$

$$5. \int_1^2 \int_0^{y^{3/2}} \frac{x}{y^2} dx dy$$

$$\begin{aligned}
 \text{Sol. } \int_1^2 \int_0^{y^{3/2}} \frac{x}{y^2} dx dy &= \int_1^2 \frac{1}{y^2} \left(\left[\frac{x^2}{2} \right]_0^{y^{3/2}} \right) dy = \int_1^2 \frac{1}{y^2} \left(\frac{y^3}{2} - \frac{0}{2} \right) dy \\
 &= \frac{1}{2} \int_1^2 y dy = \frac{1}{2} \left[\frac{y^2}{2} \right]_1^2 = \frac{1}{4} \left(4 - \frac{1}{2} \right) = \frac{3}{4}.
 \end{aligned}$$

$$6. \int_0^1 \int_x^{\sqrt{x}} (y + y^3) dy dx$$

$$\begin{aligned}
 \text{Sol. } \int_0^1 \int_x^{\sqrt{x}} (y + y^3) dy dx &= \int_0^1 \left(\left[\frac{y^2}{2} + \frac{y^4}{4} \right]_x^{\sqrt{x}} \right) dx \\
 &= \frac{1}{2} \int_0^1 \left[x - x^2 + \frac{x^2}{2} - \frac{x^4}{2} \right] dx \\
 &= \frac{1}{2} \int_0^1 \left[x - \frac{x^2}{2} - \frac{x^4}{2} \right] dx \\
 &= \frac{1}{2} \left[\frac{x^2}{2} - \frac{x^3}{6} - \frac{x^5}{10} \right]_0^1 = \frac{1}{2} \left(\frac{1}{2} - \frac{1}{6} - \frac{1}{10} \right) = \frac{7}{60}.
 \end{aligned}$$

$$7. \int_0^1 \int_0^{x^2} x e^y dy dx$$

$$\begin{aligned}
 \text{Sol. } \int_0^1 \int_0^{x^2} x e^y dy dx &= \int_0^1 \left(x [e^y]_0^{x^2} \right) dx = \int_0^1 [x (e^{x^2} - e^0)] dx \\
 &= \int_0^1 (x e^{x^2} - x) dx = \left[\frac{1}{2} e^{x^2} - \frac{x^2}{2} \right]_0^1 = \frac{1}{2} e - \frac{1}{2} - \frac{1}{2} e^0 \\
 &= \frac{e}{2} - 1.
 \end{aligned}$$

$$8. \int_2^4 \int_y^{8-y} y dx dy$$

$$\begin{aligned}
 \text{Sol. } \int_2^4 \int_y^{8-y} y dx dy &= \int_2^4 [yx]_y^{8-y} dy = \int_2^4 [y(8-y-y)] dy \\
 &= \int_2^4 (8y - 2y^2) dy = \left[4y^2 - \frac{2}{3}y^3 \right]_2^4 \\
 &= 4 \times 16 - \frac{2}{3} \times 64 - 4 \times 4 + \frac{2}{3} \times 8 = 48 - \frac{112}{3} = \frac{32}{3}.
 \end{aligned}$$

9.
$$\int_0^4 \int_{y/2}^2 e^{x^2} dx dy$$

Sol. The region of integration is bounded by $0 \leq y \leq 4$, $x = \frac{y}{2}$ and $x = 2$.

This region is also enclosed by $0 \leq x \leq 2$, $y = 0$ and $y = 2x$. The given integral is

$$\begin{aligned} &= \int_0^2 \int_0^{2x} e^{x^2} dy dx = \int_0^2 e^{x^2} [y]_0^{2x} dx \\ &= \int_0^2 2xe^{x^2} dx = [e^{x^2}]_0^2 = e^4 - 1. \end{aligned}$$

10.
$$\int_0^2 \int_{y^2}^4 y \cos x^2 dx dy$$

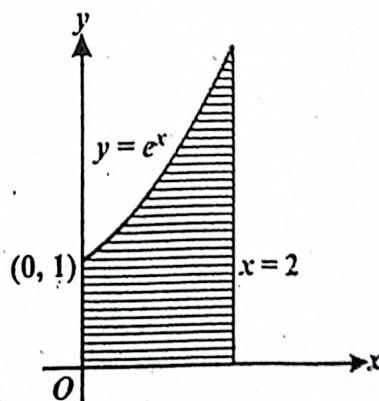
Sol. We change the order of integration. The region of integration is $0 \leq y \leq 2$, $y^2 \leq x \leq 4$. This is equivalent to $0 \leq x \leq 4$, $0 \leq y \leq \sqrt{x}$. The given integral equals

$$\begin{aligned} \int_0^4 \int_0^{\sqrt{x}} y \cos x^2 dy dx &= \frac{1}{2} \int_0^4 [y^2]_0^{\sqrt{x}} \cos x^2 dx = \frac{1}{2} \int_0^4 x \cos x^2 dx \\ &= \frac{1}{4} [\sin x^2]_0^4 = \frac{1}{4} \sin 16. \end{aligned}$$

11. $\int_D \int dy dx$ and $\int_D \int dx dy$, where D is the region bounded by the y -axis, the line $x = 2$ and the curve $y = e^x$.

Sol.
$$\begin{aligned} \int_D \int dy dx &= \int_0^2 \int_1^{e^x} dy dx = \int_0^2 (e^x - 1) dx \\ &= [(e^x - x)]_0^2 = e^2 - 3 \end{aligned}$$

$$\int_D \int dx dy = \int_1^{e^2} \int_{\ln y}^2 dx dy$$



$$= \int_1^{e^2} (2 - \ln y) dy = [2y - y \ln y + y]_1^{e^2} = e^2 - 3.$$

$$12. \int_0^{\pi/2} \int_0^2 r^2 \cos \theta dr d\theta$$

$$\begin{aligned} \text{Sol. } \int_0^{\pi/2} \int_0^2 r^2 \cos \theta dr d\theta &= \int_0^{\pi/2} \cos \theta \left[\frac{r^3}{3} \right]_0^2 d\theta = \int_0^{\pi/2} \frac{8}{3} \cos \theta d\theta \\ &= \frac{8}{3} [\sin \theta]_0^{\pi/2} = \frac{8}{3} (1 - 0) = \frac{8}{3}. \end{aligned}$$

$$13. \int_0^{2\pi} \int_0^{1-\cos \theta} r^3 \cos^2 \theta dr d\theta$$

$$\begin{aligned} \text{Sol. } \int_0^{2\pi} \int_0^{1-\cos \theta} r^3 \cos^2 \theta dr d\theta &= \int_0^{2\pi} \cos^2 \theta \left[\frac{r^4}{4} \right]_0^{1-\cos \theta} d\theta \\ &= \int_0^{2\pi} \frac{1}{4} [\cos^2 \theta (1 - \cos \theta)^4] d\theta \\ &= \frac{1}{4} \int_0^{2\pi} \cos^2 \theta [1 - 4 \cos \theta + 6 \cos^2 \theta - 4 \cos^3 \theta + \cos^4 \theta] d\theta \\ &= \frac{1}{4} \int_0^{2\pi} [\cos^2 \theta - 4 \cos^3 \theta + 6 \cos^4 \theta - 4 \cos^5 \theta + \cos^6 \theta] d\theta \\ &= \frac{1}{4} \cdot 2 \int_0^{\pi} [\cos^2 \theta - 4 \cos^3 \theta + 6 \cos^4 \theta - 4 \cos^5 \theta + \cos^6 \theta] d\theta \\ &= \frac{1}{2} \int_0^{\pi} \cos^2 \theta d\theta - 2 \int_0^{\pi} \cos^3 \theta d\theta + 3 \int_0^{\pi} \cos^4 \theta d\theta - 2 \int_0^{\pi} \cos^5 \theta d\theta + \frac{1}{2} \int_0^{\pi} \cos^6 \theta d\theta \\ &= \frac{1}{2} \cdot 2 \int_0^{\pi/2} \cos^2 \theta d\theta - 2(0) + 3 \cdot 2 \int_0^{\pi/2} \cos^4 \theta d\theta - 2(0) + \frac{1}{2} \cdot 2 \int_0^{\pi/2} \cos^6 \theta d\theta \end{aligned}$$

$$\begin{aligned}
 &= \int_0^{\pi/2} \cos^2 \theta d\theta + 6 \int_0^{\pi/2} \cos^4 \theta d\theta + \int_0^{\pi/2} \cos^6 \theta d\theta \\
 &= \frac{1}{2} \frac{\pi}{2} + 6 \frac{3 \cdot 1}{4 \cdot 2} \frac{\pi}{2} + \frac{5 \cdot 3 \cdot 1}{6 \cdot 4 \cdot 2} \frac{\pi}{2}, \text{ by Wallis formula} \\
 &= \left(\frac{1}{2} + \frac{9}{4} + \frac{15}{48} \right) \frac{\pi}{2} = \left(\frac{24 + 108 + 15}{48} \right) \frac{\pi}{2} = \frac{49}{32} \pi.
 \end{aligned}$$

14. $\int_D \int e^{-(x^2 + y^2)} dx dy$, where D is the region in the first quadrant bounded by the circles $x^2 + y^2 = 1$ and $x^2 + y^2 = 4$.

Sol. Changing into polar coordinates, the given integral is

$$\begin{aligned}
 &= \int_0^{\pi/2} \int_1^2 e^{-r^2} r dr d\theta = \int_0^{\pi/2} \left[-\frac{1}{2} e^{-r^2} \right]_1^2 d\theta \\
 &= \frac{1}{2} (e^{-1} - e^{-4}) \left(\frac{\pi}{2} \right) = \frac{\pi}{4} (e^{-1} - e^{-4}).
 \end{aligned}$$

15. $\int_D \int \frac{dx dy}{1 + x^2 + y^2}$, where D is the closed disc of radius a with centre at the origin.

Sol. Changing into polar coordinates, the given integral is

$$\begin{aligned}
 &= \int_0^{2\pi} \int_0^a \frac{r dr d\theta}{1 + r^2} = \int_0^{2\pi} \left[\frac{1}{2} \ln(1 + r^2) \right]_0^a d\theta \\
 &= \frac{1}{2} \ln(1 + a^2) \int_0^{2\pi} d\theta = \pi \ln(1 + a^2).
 \end{aligned}$$

16. $\int_D \int \frac{x^2}{(x^2 + y^2)^2} dA$, where D is the region in the first quadrant

bounded by the circles $x^2 + y^2 = a^2$, $x^2 + y^2 = b^2$, $0 < a < b$.

Sol. Changing into polar coordinates, the given integral is

$$= \int_0^{\pi/2} \int_a^b \frac{r^2 \cos^2 \theta}{r^4} r dr d\theta$$

$$= \int_0^{\pi/2} \cos^2 \theta [\ln r]_a^b d\theta = \ln \frac{b}{a} \int_0^{\pi/2} \cos^2 \theta d\theta = \frac{\pi}{4} \ln \frac{b}{a}.$$

$$17. \int_{-a}^a \int_0^{\sqrt{a^2-x^2}} (x^2+y^2)^{3/2} dy dx.$$

Sol. Changing into polar coordinates, the given integral is

$$= \int_0^{\pi} \int_0^a r^3 \cdot r dr d\theta = \int_0^{\pi} \left[\frac{r^5}{5} \right]_0^a d\theta = \frac{a^5}{5} \int_0^{\pi} d\theta = \frac{\pi a^5}{5}.$$

$$18. \int_0^1 \int_0^{\sqrt{1-y^2}} \sin(x^2+y^2) dx dy$$

Sol. Changing into polar coordinates, we have the integral

$$\begin{aligned} &= \int_0^{\pi/2} \int_0^1 (\sin r^2) r dr d\theta = \int_0^{\pi/2} \left[-\frac{1}{2} \cos r^2 \right]_0^1 d\theta \\ &= \frac{1}{2} (1 - \cos 1) \int_0^{\pi/2} d\theta = \frac{\pi}{4} (1 - \cos 1). \end{aligned}$$

$$19. \int_0^4 \int_0^{\sqrt{4y-y^2}} (x^2+y^2) dx dy$$

Sol. The region of integration is bounded by

$$0 \leq x \leq \sqrt{4y-y^2} \text{ and } 0 \leq y \leq 4$$

Now $x = \sqrt{4y-y^2}$ is the circle $x^2 + y^2 - 4y = 0$

or $x^2 + y^2 = 4y$. In polar coordinates this takes the form

$$r^2 = 4r \sin \theta, \quad \text{or} \quad r = 4 \sin \theta$$

On changing into polar coordinates, the given integral is

$$= \int_0^{\pi/2} \int_0^{4 \sin \theta} r^2 \cdot r dr d\theta = \int_0^{\pi/2} 64 \sin^4 \theta d\theta$$

$$= 64 \cdot \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2}, \text{ (using Wallis formula) } = 12\pi.$$

20. (a). Let D_a be the region bounded by the circle $x^2 + y^2 = a^2$.

$$\text{Define } \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-(x^2+y^2)} dx dy = \lim_{a \rightarrow \infty} \int_{D_a} e^{-(x^2+y^2)} dx dy.$$

Evaluate this improper integral.

Sol.(a). Changing into polar coordinates, the given integral

$$\begin{aligned} I &= \lim_{a \rightarrow \infty} \int_0^{2\pi} \int_0^a e^{-r^2} r dr d\theta = \lim_{a \rightarrow \infty} \int_0^{2\pi} \left[-\frac{1}{2} e^{-r^2} \right]_0^a d\theta \\ &= \lim_{a \rightarrow \infty} \int_0^{2\pi} \frac{1}{2} (1 - e^{-a^2}) d\theta = \lim_{a \rightarrow \infty} \left[\frac{1}{2} \theta - \frac{1}{2} e^{-a^2} \theta \right]_0^{2\pi} \\ &= \pi - \lim_{a \rightarrow \infty} \frac{\pi}{e^{a^2}} = \pi. \end{aligned}$$

20. (b). Use part (a) to prove that $\int_0^{\infty} e^{-x^2} dx = \frac{1}{2} \sqrt{\pi}$.

$$\begin{aligned} \text{Sol. } \int_{-a}^a \int_{-a}^a e^{-(x^2+y^2)} dy dx &= \int_{-a}^a e^{-y^2} \left(\int_{-a}^a e^{-x^2} dx \right) dy \\ &= \left(\int_{-a}^a e^{-x^2} dx \right) \left(\int_{-a}^a e^{-y^2} dy \right) = \left(\int_{-a}^a e^{-x^2} dx \right)^2 = 4 \left(\int_0^a e^{-x^2} dx \right)^2 \end{aligned}$$

Letting $a \rightarrow \infty$, we have

$$\begin{aligned} &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-(x^2+y^2)} dx dy \\ &= 4 \left(\int_0^{\infty} e^{-x^2} dx \right)^2 = \pi, \text{ from Part (a) above.} \end{aligned}$$

$$\text{Therefore, } \int_0^{\infty} e^{-x^2} dx = \frac{1}{2} \sqrt{\pi}.$$